



**Report On
Resource Adequacy Plan
For the State of
Tamil Nadu
(2024-25 to 2034-35)**

**Government of India
Ministry of Power
Central Electricity Authority**

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Executive Summary

As per Rule 16 of the Electricity (Amendment) Rules 2022, Ministry of Power has notified Resource Adequacy guidelines. As per the Resource Adequacy (RA) Guidelines, Central Electricity Authority is entrusted to prepare Long Term-National Resource Adequacy Plan (LT-NRAP). Further Distribution Utility need to carry out LT-DRAP (Long-term Distribution Licensee Resource Adequacy Plan) to meet the utility peak and energy requirement reliably.

As per the Resource Adequacy Guidelines, each Distribution licensee shall undertake a Resource Adequacy Plan (RAP) for a 10-year horizon (Long-term Distribution Licensee Resource Adequacy Plan (LT-DRAP)) to meet their own peak and electrical energy requirement. The plan shall be vetted/validated by Central Electricity Authority for leveraging the benefit of national level optimization for the Distribution licensees. The LT-DRAP shall be carried out by the distribution licensees on an annual rolling basis considering the contracted capacity as a part of the system and shall optimize for additional capacity required.

Generation Capacity expansion study have been carried out considering existing capacity, planned capacity & capacity required to fulfil the Renewable Purchase Obligations (RPO) as per the MoP order issued in Oct 2023. It was observed that the total unserved energy in the year 2034-35 is expected to be about 45,587 MU which is about 18.3% of annual energy during the year 2034-35 and primarily observed in the months of February and March.

Therefore, to meet the demand in least cost , generation capacity expansion studies for the period 2025-26 to 2034-35, for the State of Tamil Nadu have been carried with an objective to minimize the total system cost of generation including the cost of anticipated future investments while fulfilling all the technical/financial constraints associated with various power generation technologies to meet the demand reliably. Then, Reliability study has been carried out to determine the probability of Unmet demand and hours by implementing the variation in demand, RE (Solar, Wind) and forced outage of thermal generators (Coal, gas) etc. Thus, the Resource adequacy studies have projected likely optimal capacity mix for future years till 2034-35 which is able to meet anticipated demand reliably at every instance while meeting Renewable Purchase Obligations (RPO).

1.0 Introduction

Ministry of Power has notified Electricity (Amendment) Rules, 2022 in December 2022. Rule 16 (I) of the said rules stipulates that “A guideline for assessment of resource adequacy during the generation planning stage (one year or beyond) as well as during the operational planning stage (up to one year) shall be issued by the Central Government in consultation with the Authority”. Accordingly, the Resource Adequacy Guidelines were notified in June 2023 by the Ministry of Power in consultation with Central Electricity Authority.

Resource Adequacy is generally defined as a mechanism to ensure that there is an adequate supply of generation resources to serve expected demand reliably at the least cost. A key aspect of resource adequacy planning is to ensure that adequate generation capacities are available round-the-clock to reliably serve demand under various scenarios. This naturally translates into the need for ensuring adequate reserve margin, which could cater to varying levels of demand and supply conditions in the grid. In the wake of high RE generation, it is important to understand demand-supply situation in the grid precisely due to high seasonality and intermittency in RE generation. Resource Adequacy exercise may also help in assessment of capacity requirement to be tied up or contracted on long term, medium term, and short-term basis.

Further, Ministry of Power vide notification dated 20th October 2023 had notified the RPO trajectory for the states. Based on the trajectory specified the hydro, wind and other (solar, biomass etc.) RPO quantum in million units (MUs) has been calculated to find additional quantum of renewable capacity that the states have to contract in addition to its existing/planned capacity to meet their RPO targets.

Resource Adequacy studies has been carried out for Tamil Nadu based on the inputs received from SLDC (Tamil Nadu) while fulfilling RPO trajectory. The study suggests the optimal resource mix till 2034-35 taking into account all technical and financial parameters associated with capacities. The study optimizes power purchase on a long-term basis while evaluating resource adequacy for meeting the demand 24 X 7 considering variation in demand, RE generation and forced outages of thermal capacities. The study has also assessed the requirement of Planning Reserve margin for Tamil Nadu for catering to above highlighted uncertainties so that demand can be met reliably throughout the year.

2.0 Tamil Nadu RA Study

2.1 Present Power Scenario in Tamil Nadu

The current Power Supply Position for Tamil Nadu is shown in the Table 1 below:

Table 1 Power Supply Position

Power Supply Position						
Year	Energy required (MU)	Energy Supplied (MU)	Gap(MU)	Peak Demand (MW)	Peak Met (MW)	Demand Not Met (MW)
2021-22	1,09,816	1,09,798	18	16,891	16,891	0
2022-23	1,14,798	1,14,722	76	17,729	17,729	0
2023-24	1,26,163	1,26,151	12	19,045	19,045	0

As of March 2024, the total contracted capacity for Tamil Nadu is 36593 MW. Out of the total contracted capacity (CC), the share of non-fossil fuel-based CC is 61%.

The fuel-wise contracted capacity as on March 2024 is given in Table 2 and Figure 1 below:

Table 2 Fuel-wise Contracted Capacity (in MW) as on March 2024

Technology	Contracted Capacity (MW)	%
COAL	13376	37%
GAS	739	2%
NUCLEAR	1755	5%
BIOMASS	891	2%
HYDRO	1864	5%
WIND	9015	25%
SOLAR	7896	22%
PSP	400	1%
Hydro DRE	58	0%
Solar DRE Rooftop	599	2%
Total	36593	100%

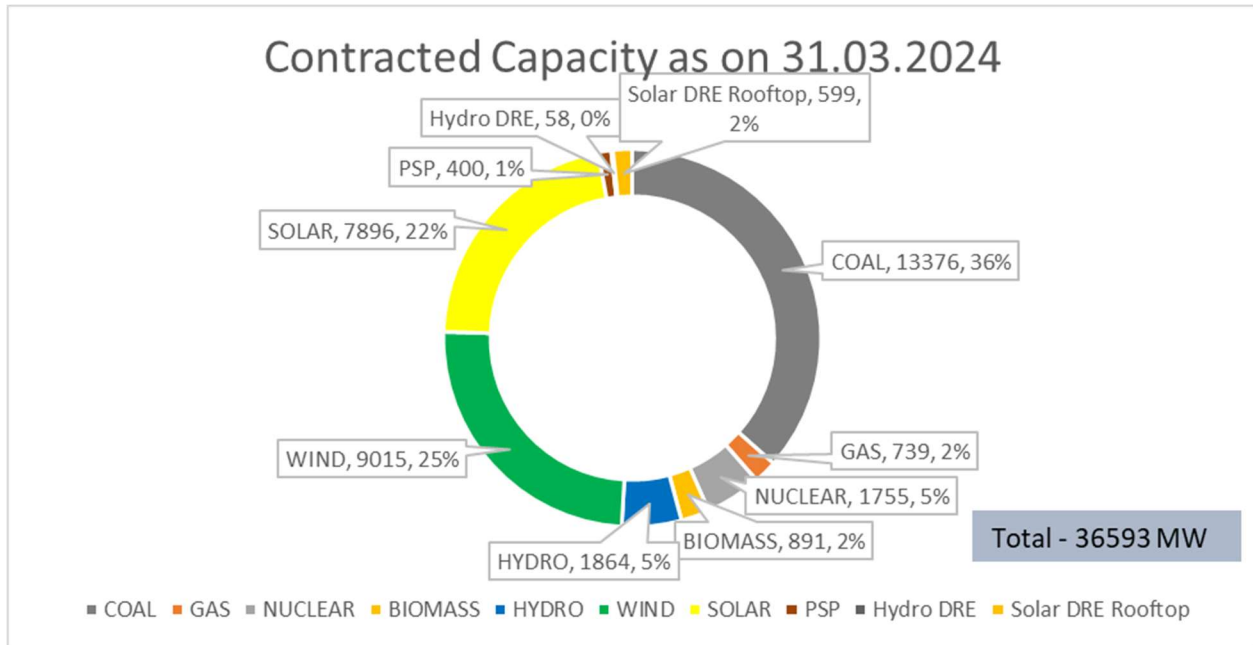


Figure 1 Fuel-wise Contracted Capacity (in MW) as of Mar 2024

2.2 Present Demand Analysis (2023-24)

The hourly demand pattern of 2023-24 was analyzed (as shown in Figure 2 and Figure 3), and it was observed that the peak demand for Tamil Nadu occurs in the months of March and April. The Demand Pattern of Tamil Nadu indicates that the state has a higher Day peak in comparison to its Night Peak, with the daily peak mostly occurring during the 11:00 Hrs to 15:00 Hrs. Further, the demand pattern remains almost flat during the noon hours.

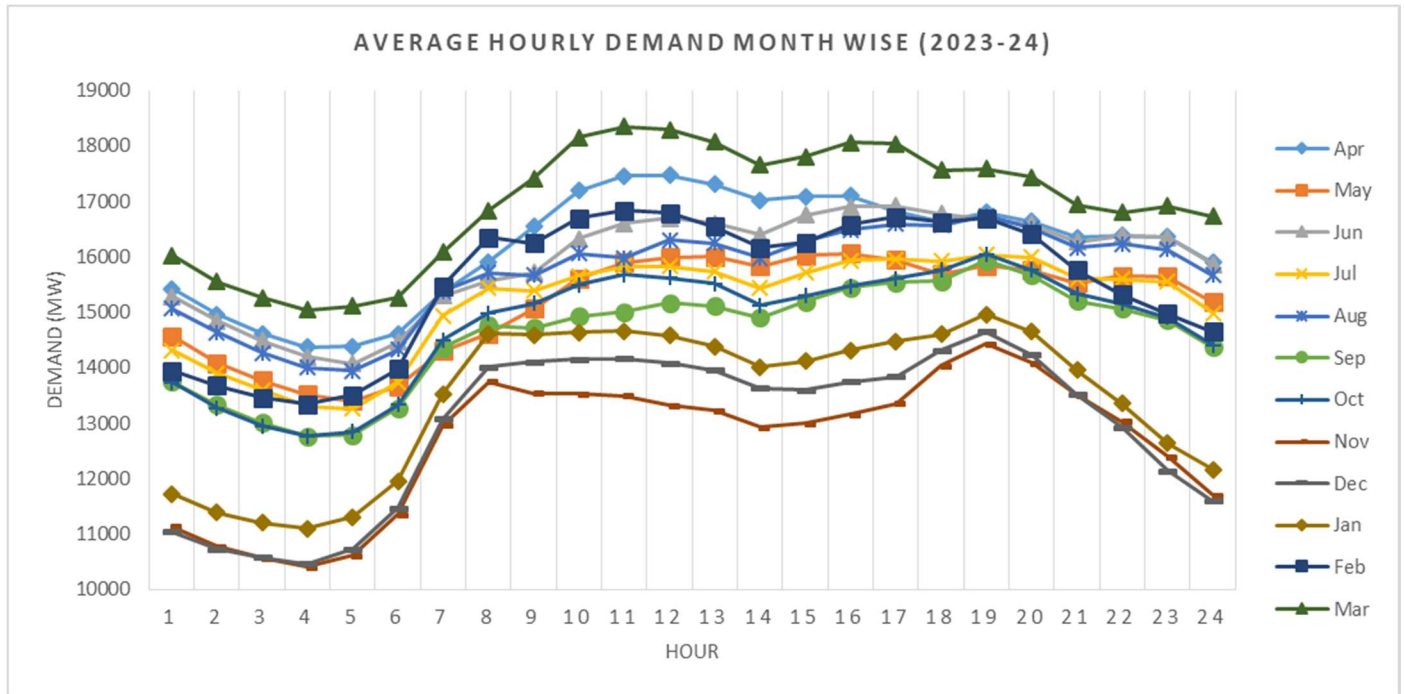


Figure 2 Average Hourly Demand Variation (Month-wise) of Tamil Nadu for 2023-24

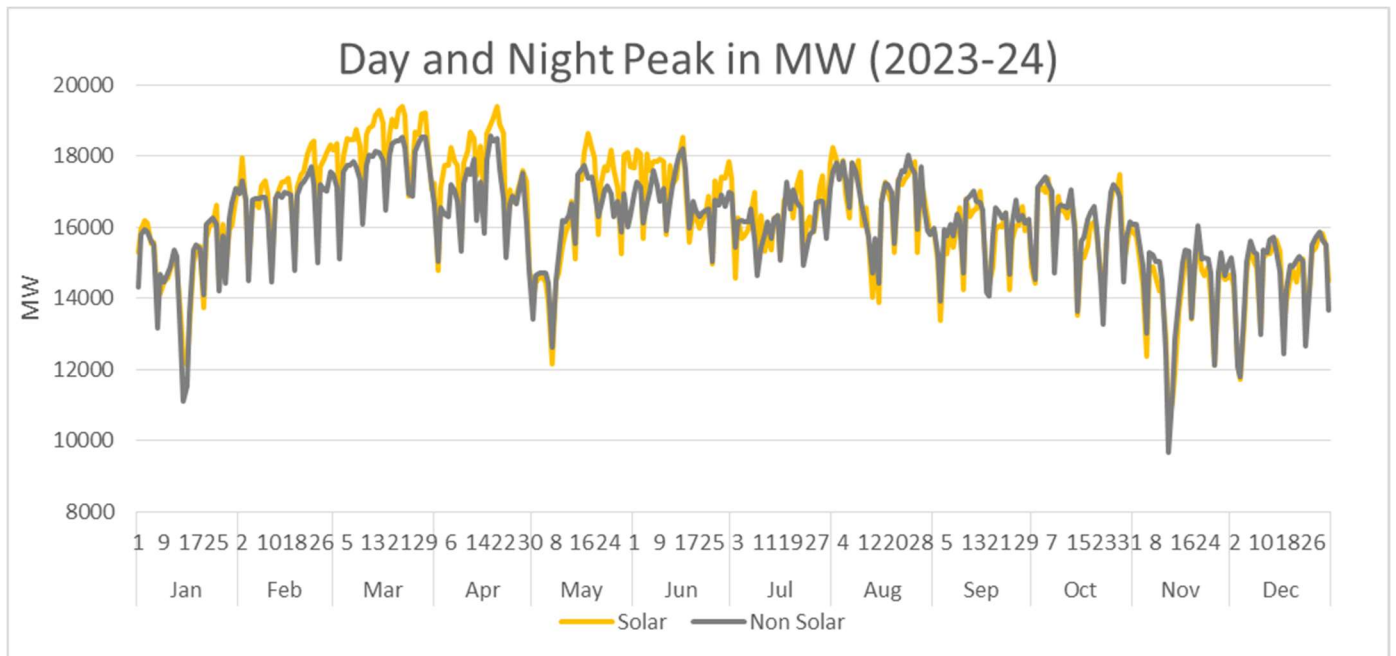


Figure 3 Day and Night Peak in MW of Tamil Nadu (2023-24)

The hourly demand pattern of 2023-24 was analysed (Figure 4) to find out the number of occurrences of the peak and near peak demand. Such instances are critical for study purpose as it is necessary to ensure resource adequacy during such instances with an optimal mix of long-term, medium-term and short-term contracts.

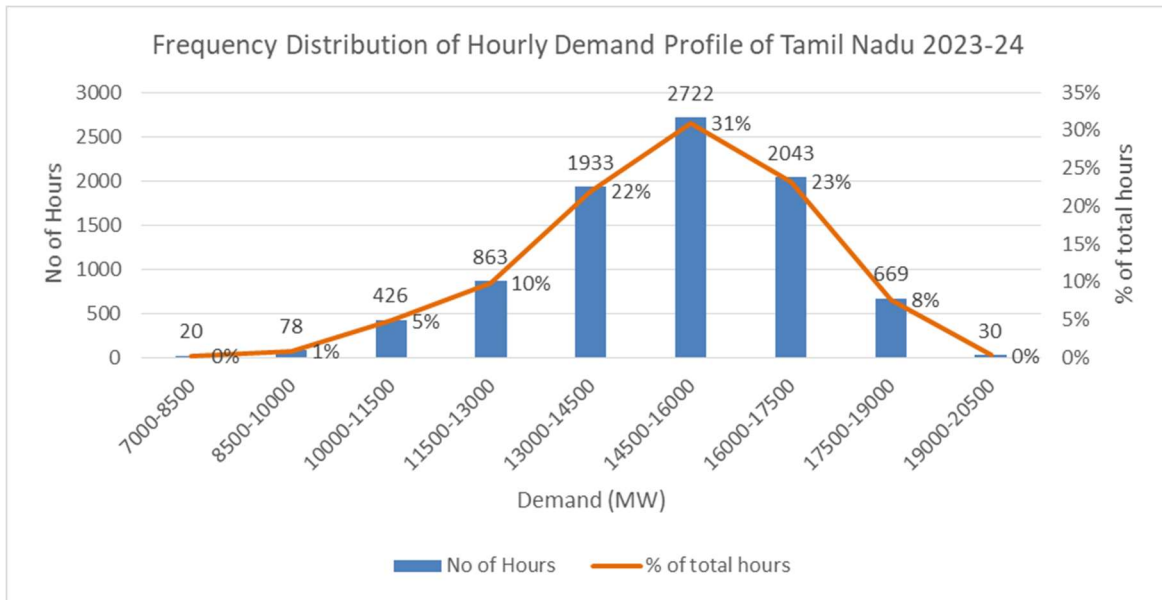


Figure 4 Frequency Distribution of Hourly Demand Profile of Tamil Nadu 2023-24

3.0 Inputs/Assumptions for the Study

- i) Peak and Energy Demand for the State of Tamil Nadu, as furnished by the utility, were compared with the 20th EPS (Electric Power Survey) projections (Table 3). The Demand estimation by 20th EPS was found to be lower than projected by Tamil Nadu as shown in Figure 5. Therefore, the Studies have been carried out using the projections received from Tamil Nadu.

Table 3 Future Demand Projection by Tamil Nadu

	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32	2032-33	2033-34	2034-35
Energy Projections (MU)	130223 (126163*)	142441	152678	162726	172679	182401	193232	203598	214582	225898	237559	249580
Year on Year Growth		9.4%	7.2%	6.6%	6.1%	5.6%	5.9%	5.4%	5.4%	5.3%	5.2%	5.1%
Peak Demand Projections (MW)	19409 (19045*)	20701	21959	23314	24680	26046	27541	29027	30567	32159	33805	35507
Year on Year Growth		6.7%	6.1%	6.2%	5.9%	5.5%	5.7%	5.4%	5.3%	5.2%	5.1%	5.0%

*Actual Demand Observed in FY 2023-24

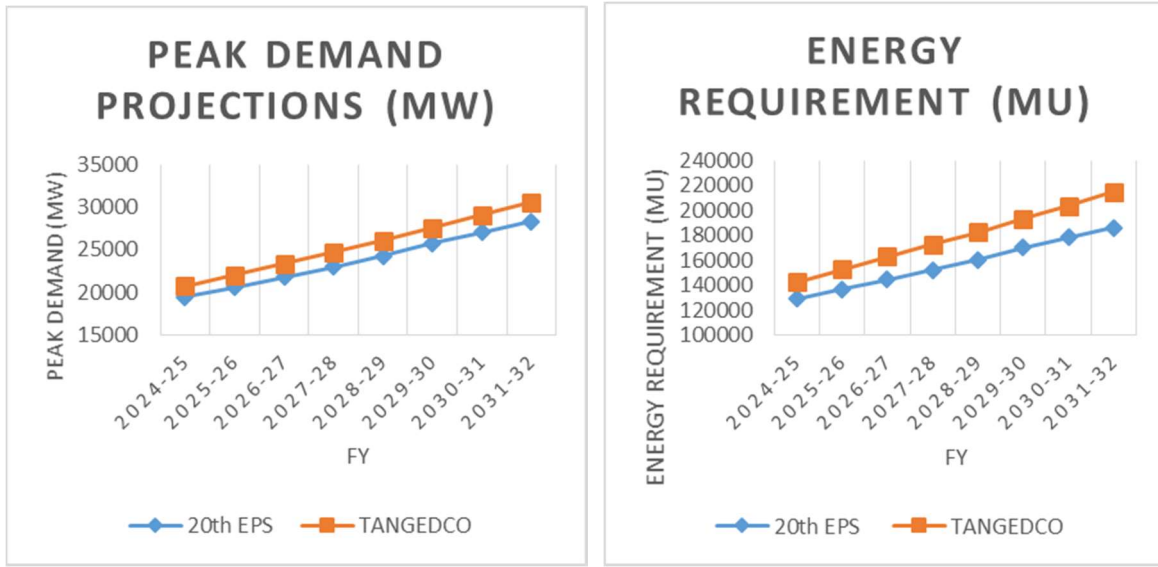


Figure 5 Comparison of Energy Requirement and peak Demand Projections of 20th EPS vs furnished by Tamil Nadu state

- ii) Future hourly demand profile till the year 2034-35 has been projected, taking the demand profile of FY 2023-24 as a base profile.
- iii) The actual hourly solar and wind generation profiles have been referred to as per the inputs received from SLDC, Tamil Nadu.
- iv) Capital cost of candidate plants for Coal, Wind, Solar, Battery and PSP, considered as investment options for the studies, have been referred from National Electricity Plan 2022-32(Vol-I generation).
- v) Existing, Planned, and Expiring Capacity Due to PPA Termination/Retirement: As per the information received from Tamil Nadu as shown in Table 2, Table 4(a) and Table 4(b) respectively.

Table 4(a) Year on Year source wise planned capacity addition (MW)

FY	COAL	NUCLEAR	BIOMASS	HYDRO	WIND	SOLAR	PSP	Total
2024/25	2120	152	0	0	500	4100	0	6872
2025/26	0	0	75	20	500	1800	500	2895
2026/27	1320	1100	0	0	500	1500	0	4420
2027/28	462	0	0	0	500	1700	0	2662
2028/29	2082	0	0	0	500	2000	0	4582
2029/30	500	1000	0	0	2000	2300	0	5800
2030/31	0	0	0	0	1000	1000	0	2000
2031/32	0	0	0	0	1000	1000	0	2000
2032/33	0	0	0	0	1000	1000	0	2000
2033/34	0	0	0	0	1000	1000	1000	3000
2034/35	0	0	0	0	1000	1000	0	2000
TOTAL	6484	2252	75	20	9500	18400	1500	38231

Table 4(b) Year on Year source wise capacity expiration (MW)

FY	COAL	GAS	NUCLEAR	Total
2024/25	205	0	0	205
2025/26	150	0	0	150
2026/27	0	0	0	0
2027/28	0	0	0	0
2028/29	2417	0	1178	3595
2029/30	708	0	0	708
2030/31	0	0	0	0
2031/32	0	331	0	331
2032/33	0	0	0	0
2033/34	250	0	0	250
2034/35	633	0	0	633
TOTAL	4363	331	1178	5871

- vi) **Renewable Purchase Obligation (RPO) trajectory:** Ministry of Power gazette notification dated 20th October 2023 had notified the source-wise minimum share of consumption of non-fossil sources (renewable energy) by designated consumers till the year 2029-30. In view of the country's energy transition goals as well as the long-term net zero target of 2070, it is estimated that the share of RE generation in the generation mix will continue to increase beyond 2029-30. Therefore, the RPO trajectory is assumed to rise steadily beyond 2029-30, and hence, RPO targets till 2034-35 are considered as given below in Table 5:

Table 5 Renewable Purchase Obligation (RPO) trajectory as per MoP order*

Sl. No.	Year	Wind renewable energy	Hydro renewable energy	Other renewable energy	Distributed renewable energy	Total renewable energy
(1)	(2)	(3)	(4)	(6)	(5)	(7)
1.	2024-25	0.67%	0.38%	27.35%	1.5%	29.91%
2.	2025-26	1.45%	1.22%	28.24%	2.1%	33.01%
3.	2026-27	1.97%	1.34%	29.94%	2.7%	35.95%
4.	2027-28	2.45%	1.42%	31.64%	3.3%	38.81%
5.	2028-29	2.95%	1.42%	33.10%	3.9%	41.36%
6.	2029-30	3.48%	1.33%	34.02%	4.5%	43.33%
7.	2030-31	40.50%			5.0%	45.50%
8.	2031-32	41.50%			5.5%	47.00%
9.	2032-33	42.30%			6.0%	48.30%
10.	2033-34	43.00%			6.5%	49.50%

11.	2034-35	43.50%	7.0%	50.50%
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*Trajectory for RPO till 2029-30 as per MoP RPO order notified in October,2023. After 2029-30, RPO targets assumed based on anticipated RE capacity requirement on national level given in National Electricity Plan 2022-32, (Vol-I Generation)

Based on the trajectory specified, RPO quantum in million units (MUs) from hydro, wind, other (solar, biomass etc.) and distributed renewable energy (DRE) is calculated and tabulated below in Table 6:

*Table 6 Total Energy required to meet RPO (MU)**

Sl. No.	Year	Wind renewable energy (MU)	Hydro renewable energy	Other renewable energy	Distributed renewable energy	Total renewable energy
(1)	(2)	(3)	(4)	(6)	(5)	(7)
1	2024-25	954	541	38958	2137	42604
2	2025-26	2214	1863	43116	3206	50399
3	2026-27	3206	2181	48720	4394	58500
4	2027-28	4231	2452	54636	5698	67017
5	2028-29	5381	2590	60375	7114	75441
6	2029-30	6724	2570	65738	8695	83727
7	2030-31	82457			10180	92637
8	2031-32	89052			11802	100854
9	2032-33	95555			13554	109109
10	2033-34	102150			15441	117592
11	2034-35	108567			17471	126038

*Considering the fungibility aspect of RPO targets among Wind, Hydro and other RE generation

The year-wise total renewable energy met with the existing and planned capacity along with the deficit is shown in Table 7 below:

Table 7 Renewable Energy Deficit

FY	Renewable energy required to meet RPO	Renewable energy required to meet RPO	Renewable energy available	Renewable energy available	Surplus(+)/ Deficit (-)
	(MU)	(%)	(MU)	(%)	(%)
2024/25	42604	29.91%	40405	28.37%	-1.54%
2025/26	50399	33.01%	44978	29.46%	-3.55%
2026/27	58500	35.95%	47960	29.47%	-6.48%
2027/28	67017	38.81%	51278	29.70%	-9.11%
2028/29	75441	41.36%	56002	30.70%	-10.66%
2029/30	83727	43.33%	62202	32.19%	-11.14%
2030/31	92637	45.50%	66030	32.43%	-13.07%
2031/32	100854	47.00%	70095	32.67%	-14.33%
2032/33	109109	48.30%	74162	32.83%	-15.47%

2033/34	117592	49.50%	79813	33.60%	-15.90%
2034/35	126038	50.50%	84262	33.76%	-16.74%

Accordingly, the additional source wise MW requirement considering the fungibility aspects in the RPO has been estimated by the model as tabulated below in Table 8:

Table 8 As per RPO trajectory, Tamil Nadu needs to add/contract following additional capacity (MW).

FY	WIND	SOLAR	DRE	TOTAL
2024/25	500	500	715	1715
2025/26	500	500	342	1342
2026/27	500	500	185	1185
2027/28	500	500	277	1277
2028/29	500	500	364	1364
2029/30	500	500	245	1245
2030/31	500	500	169	1169
2031/32	500	500	277	1277
2032/33	500	500	279	1279
2033/34	500	500	386	1386
2034/35	500	500	498	1498
TOTAL	5500	5500	3737	14737

Thus, the future RE capacity tie-ups, besides the capacity that has already been planned, in order to meet the RPO trajectory, are 14737 MW (5500 MW Wind, 5500 MW Solar and 3737 MW DRE).

4.0 Reliability Analysis

One of the main criteria of resource adequacy studies is to determine the reliability of the system to meet the demand adequately at any instance of time. This reliability is measured via two indices (i.e.) LOLP (Loss of Load Probability) and EENS (Expected Energy Not Served). These indices have been defined in resource adequacy guidelines as below:

- **Loss of Load Probability (LOLP):** Measure of the probability that a system's load may exceed the generation and firm power contracts available to meet that load in a year. For the purpose of this study, 0.2% LOLP has been considered.
- **Expected Energy Not Served (EENS):** Expected amount of energy (MWh) that may not be served for each year within the planning period under study. It is a summation of the expected number of megawatt hours of demand that may not be served for the year. This is an energy-centric metric that considers the magnitude and duration of energy being not served, calculated in Mega Watt hours (MWh). The metric can be normalized (i.e., divided by total system load) to create a Normalized Energy Not Served (NENS) metric. For the purpose of this study, 0.05% EENS has been considered.

Monte Carlo /Stochastic simulation has been used to factor-in the uncertainty associated with various generation resources and demand. It is an approach which is used to predict the probability of a variety of outcomes when the potential for random variables is present as compared to deterministic modelling of economic dispatch model. Monte Carlo simulation helps in analysing the randomness associated with RE energy resource, demand pattern changes and forced outages of plant. A large no of random samples of these variables are simultaneously simulated to ascertain system reliability indices (i.e. Loss of load probability LOLP & Energy Not Served (ENS)) & the system robustness in case of above variation of system parameters.

Planning Reserve Margin (PRM): To meet the prescribed standard of LOLP / NENS conditions, sufficient reserve margins need to be maintained in the system for adequately addressing the demand and supply variations. Planning Reserve Margin (PRM) is the predominant metric used to ensure adequacy of generation resources in the system. PRM in a power system is expressed as a certain % of peak load forecast of the system.

4.1 Demand variation:

The hourly demand variation for consecutive years (i.e., 2022-23 and 2023-24) has been analyzed. The Demand pattern variation of 2022-23 and 2023-24 is shown below in Figure 6.

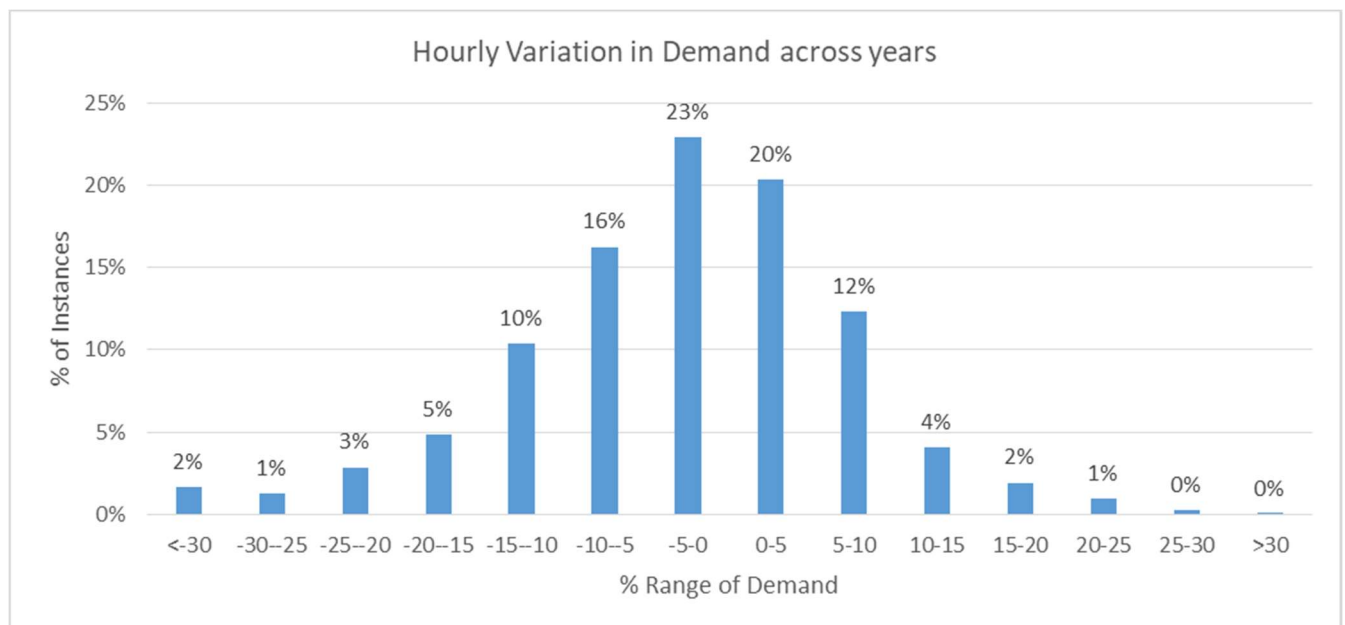


Figure 6 Hourly Variation in Demand across years

It can be observed that the hourly demand typically varies $\pm 15\%$ for 86% of instances. This variation is primarily due to temperature, weather parameters or any random outages of

transmission lines and generation units, etc. This variation has been captured in the reliability study by varying the projected hourly demand for the future years by varying $\pm 15\%$ by introducing a random variable (with normal distribution) for demand as per observed behavior over the years.

4.2 RE variation

In the Long-term capacity expansion planning studies, a particular profile for Solar and Wind Plants are considered based on the observed solar and wind generation data to determine the optimal capacity mix. However, due to intermittent nature of these sources the generation from these non-dispatchable sources may vary across years. As per the analyses carried out based on historical generation data, solar generation and wind generation has been varied by 10 % and 50% respectively to incorporate the variation in these generation sources and plan for requisite measures to mitigate such behavior.

4.3 Forced Outage of Thermal Generators

The average forced outage rate of thermal generators is typically at 10% with $\pm 5\%$ variation. The same has been incorporated in the model.

Based on these variations, reliability studies are carried out to ascertain robustness of the system. The LOLP & EENS of the system is within specified range.

5.0 Results of the study

5.1 Unserved Demand Projections

The study was carried out considering existing capacity and planned capacity only while meeting the annual RPO obligations. It was observed that the total unserved energy in the year 2034-35 is about 45587 MU. The yearly likely unserved energy with the planned capacities is given below in Figure 7.

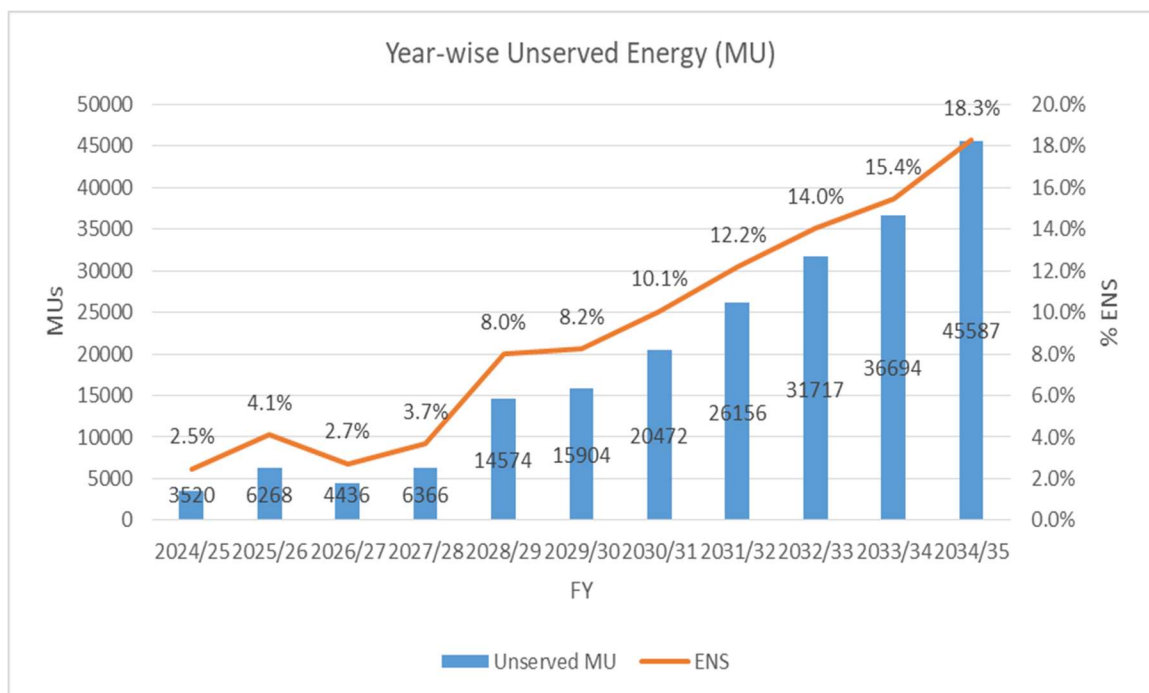


Figure 7 Yearly likely unserved energy with the planned capacities for Tamil Nadu (in MU)

The study has also analyzed the Aggregated Hourly Unserved Energy for Tamil Nadu in the year 2034-35, it can be seen from Figure 8 that the unserved energy coincides with peak demand months when the contracted capacity (present, planned and additional RE for RPO) is unable to meet the demand.

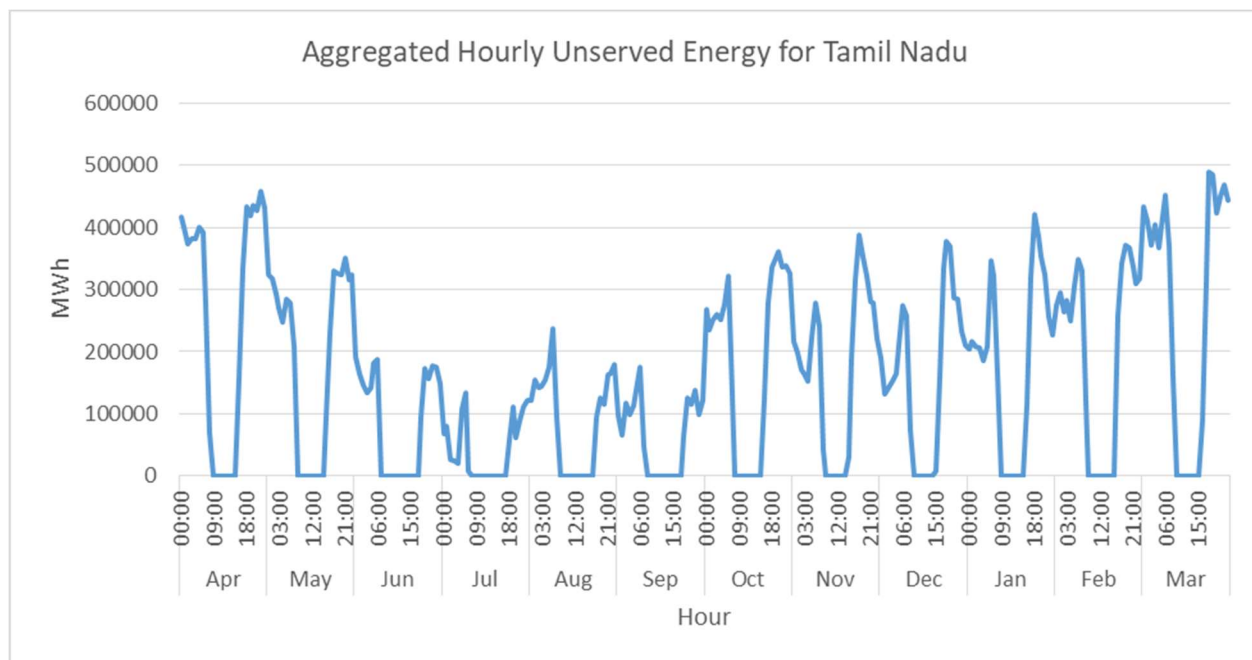


Figure 8 Aggregated Hourly Unserved Energy MWh (2034-35)

5.2 Capacity Mix Projection

The study was carried out considering existing capacity, planned capacity and additional capacity required to fulfil the RPO obligations.

The year-wise capacity projections for Tamil Nadu are given below in Table 9 and Figure 9:

Table 9 Year-wise contracted capacity projections (in MW)

	COAL	GAS	NUCLEAR	BIOMASS	HYDRO	WIND	SOLAR	Solar DRE	Hydro DRE	STORAGE	STOA
2024/25	15291	739	1906	891	1864	10015	12496	1614	58	400	3274
2025/26	15141	739	1906	966	1884	11015	14796	2457	58	900	3996
2026/27	16461	739	3006	966	1884	12015	16796	3391	58	2441	2580
2027/28	16923	739	3006	966	1884	13015	18996	4419	58	4344	2674
2028/29	17588	739	1828	966	1884	14015	21496	5533	58	5581	3310
2029/30	18380	739	2828	966	1884	16515	24296	6778	58	7581	2408
2030/31	19380	739	2828	966	1884	18015	25796	7947	58	8581	2780
2031/32	20380	408	2828	966	1884	19515	27296	9224	58	9581	2855
2032/33	21380	408	2828	966	1884	21015	28796	10603	58	10581	2184
2033/34	22130	408	2828	966	1884	22515	30296	12089	58	12581	2084
2034/35	22497	408	2828	966	1884	24015	31796	13687	58	13581	2799

The projected contracted capacity mix, year-wise is given below in the Figure 9:

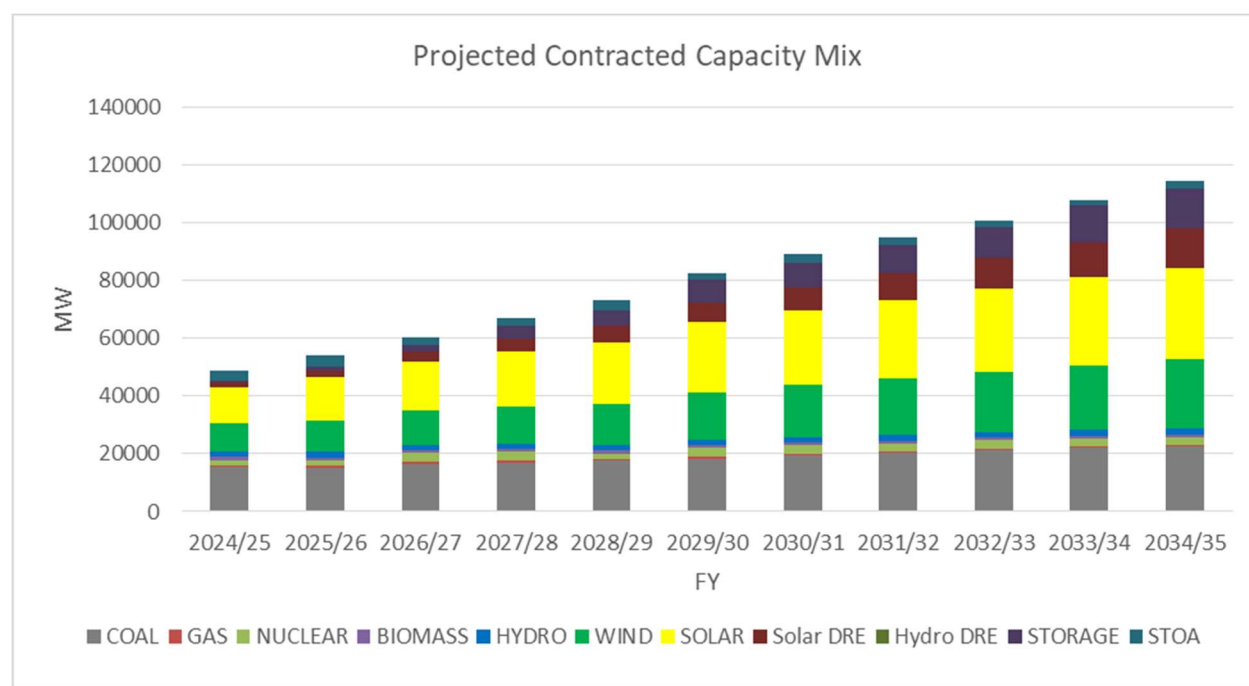


Figure 9 Projected Contracted Capacity Mix Year-wise (MW) for Tamil Nadu

As per the Resource Adequacy studies, the total projected contracted Capacity for the year 2034-35 is 98,140 MW (as shown in Figure 10) which consists of 22497 MW from Coal, 408 MW from Gas, 2828 MW from Nuclear, 966 MW from Biomass, 1884 MW from Hydro, 31796 MW from Solar, 24015 MW from Wind and 13744 MW from DRE. Additionally, 13581 MW/78122 MWh of Storage and 2799 MW from short-term/medium-term/bilateral arrangements would be required to meet the state demand reliably. This capacity shall be able to meet the projected demand with prescribed reliability criteria.

The Reliability studies have been carried out to adhere to the reliability criteria of LoLP and NENS as provided in NEP (0.2% and 0.05% respectively). The PRM for the state of Tamil Nadu has been assessed as 19% for the year 2034-35. In addition, the projected/contracted capacity fulfils the stipulated Renewable Purchase Obligation and aforementioned reliability criteria.

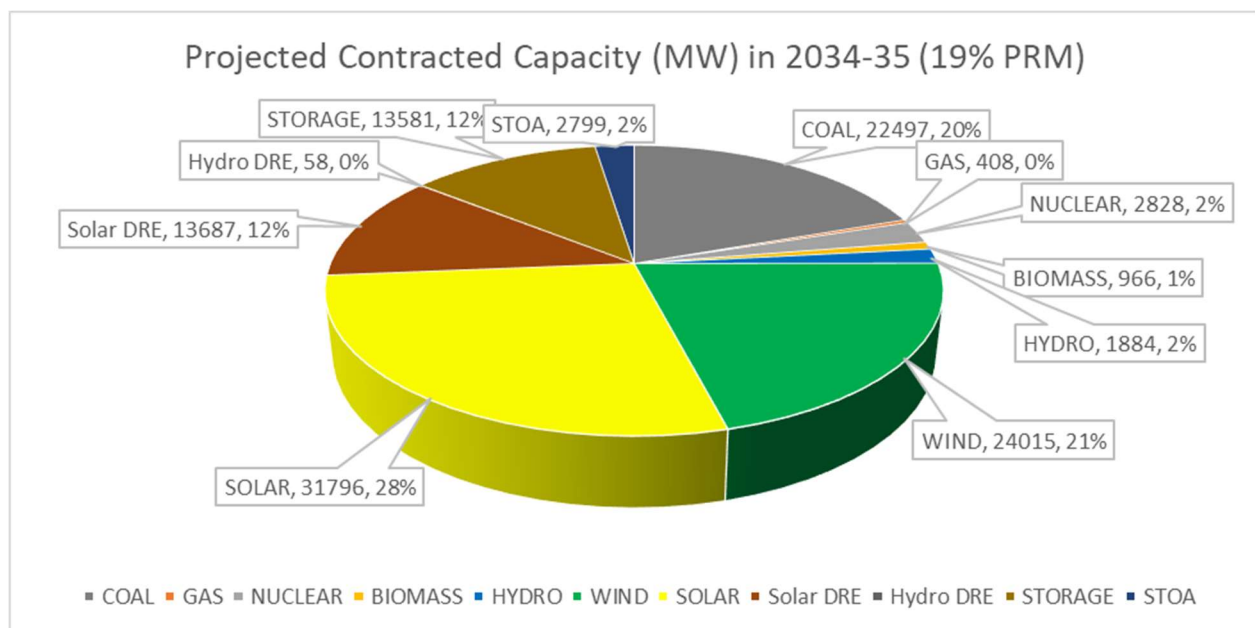


Figure 10 Projected Capacity Mix in 2034-35 with 25% PRM

The share of non-fossil fuel-based capacity in the generation mix is projected to increase to around 77% by 2034-35 from 64% in 2024-25 with higher contribution from non-fossil fuel-based capacities in alignment with RPO trajectory.

The projected generation mix for the state is shown in Figure 11 & 12 below:-

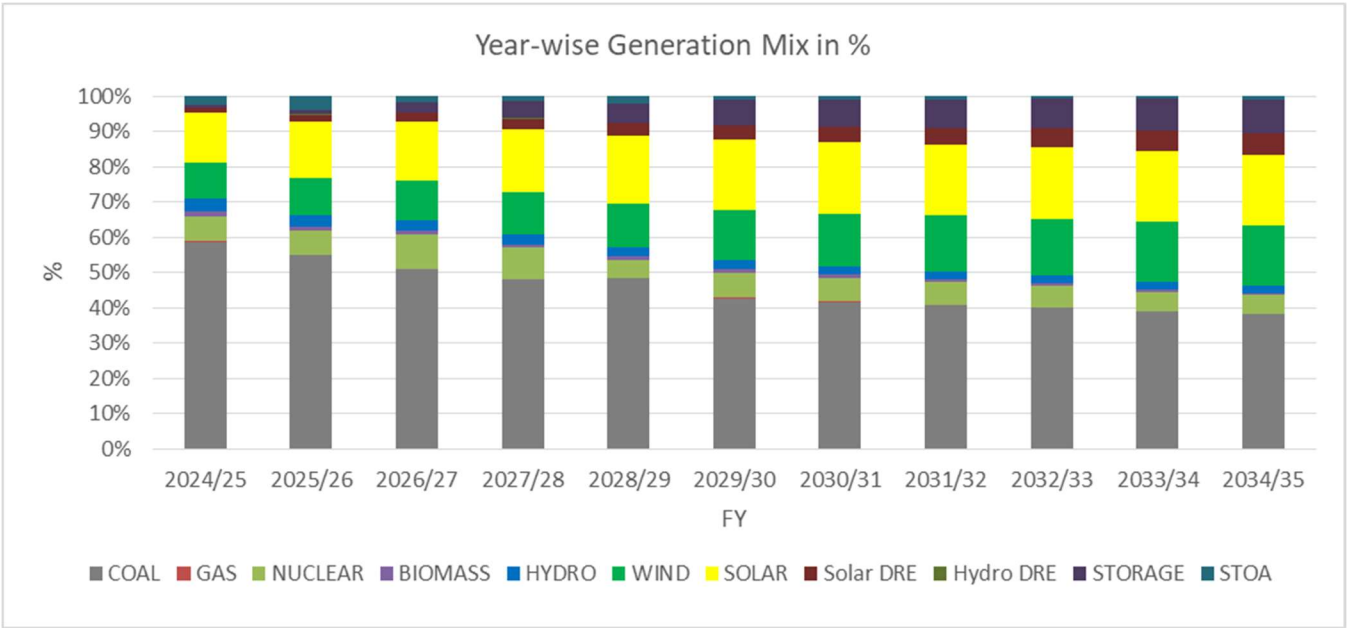


Figure 11 Projected Generation Mix (%)

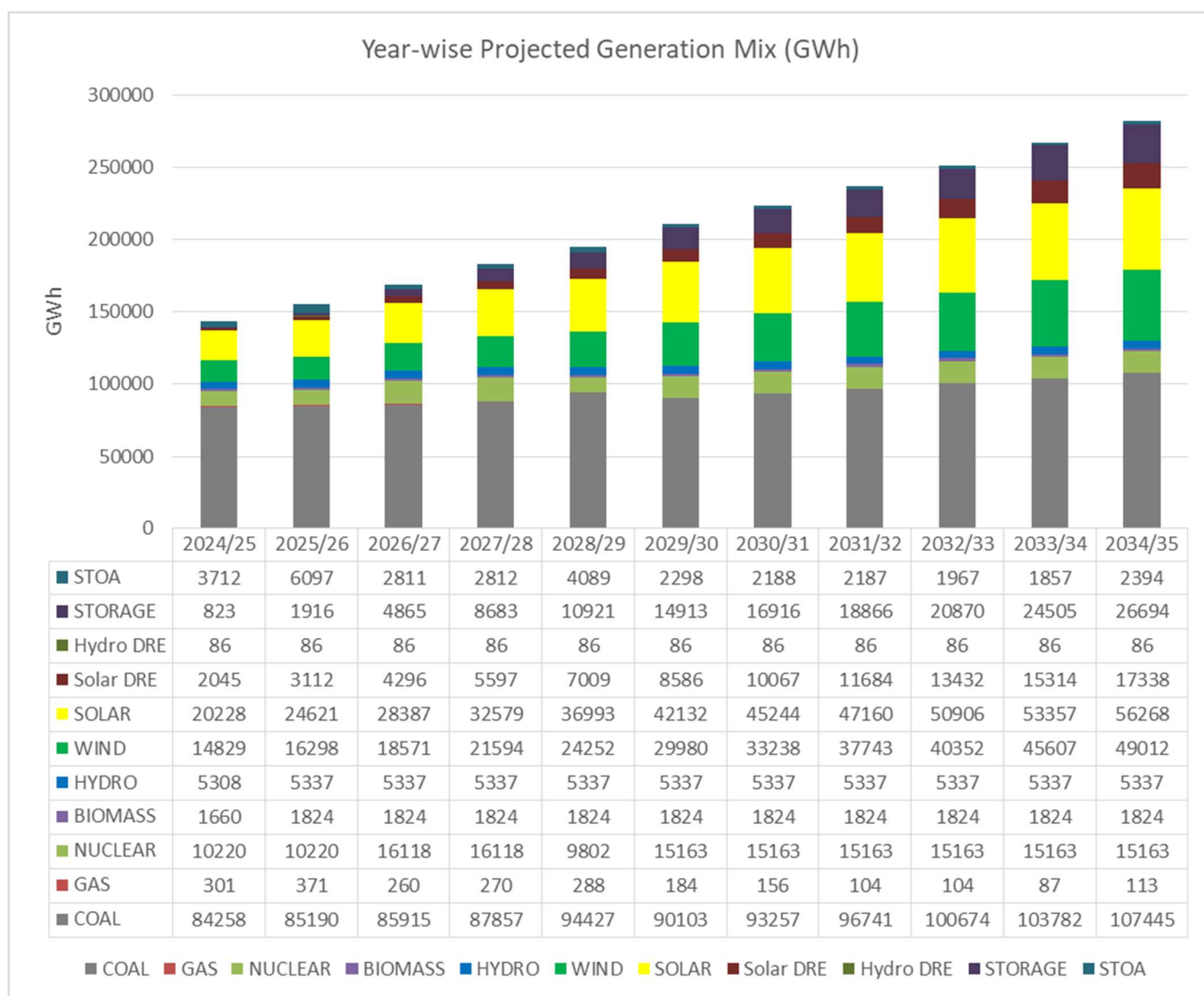


Figure 12 Year-wise projected net generation mix (in GWh)

5.3 Capacity contract requirement for future

It has been found out in the studies that Tamil Nadu needs to contract following capacities (planned and additional) per year till 2034-35 to meet its demand reliably while ensuring fulfilment of its RPO obligations as tabulated in Table 10.

FY	COAL		SOLAR		Wind		Yearly STOA
	Planned	Additional	Planned	Additional	Planned	Additional	Additional
2024/25	2120	0	4100	500	500	500	3274
2025/26	0	0	1800	500	500	500	3996
2026/27	1320	0	1500	500	500	500	2580
2027/28	462	0	1700	500	500	500	2674
2028/29	2082	1000	2000	500	500	500	3310
2029/30	500	1000	2300	500	2000	500	2408
2030/31	0	1000	1000	500	1000	500	2780
2031-32	0	1000	1000	500	1000	500	2855
2032/33	0	1000	1000	500	1000	500	2184
2033/34	0	1000	1000	500	1000	500	2084
2034/35	0	1000	1000	500	1000	500	2799
FY	Storage		DRE		NUCLEAR	Biomass	Hydro
	Planned	Additional	Planned	Additional	Planned	Planned	Planned
2024/25	0	0	300	715	152	0	0
2025/26	500	0	500	342	0	75	20
2026/27	0	1541	750	185	1100	0	0
2027/28	0	1903	750	277	0	0	0
2028/29	0	1237	750	364	0	0	0
2029/30	0	2000	1000	245	1000	0	0
2030/31	0	1000	1000	169	0	0	0
2031-32	0	1000	1000	277	0	0	0
2032/33	0	1000	1100	279	0	0	0
2033/34	1000	1000	1100	386	0	0	0
2034/35	0	1000	1100	498	0	0	0

Table 10 Year wise Capacity Addition for Tamil Nadu (in MW)

5.4 Projected Coal Capacity utilization

The coal capacity PLF is expected to remain in the range of 54%-64 % as shown below in Figure 13 for the years till 2034-35 ensuring higher absorption of higher renewable energy.

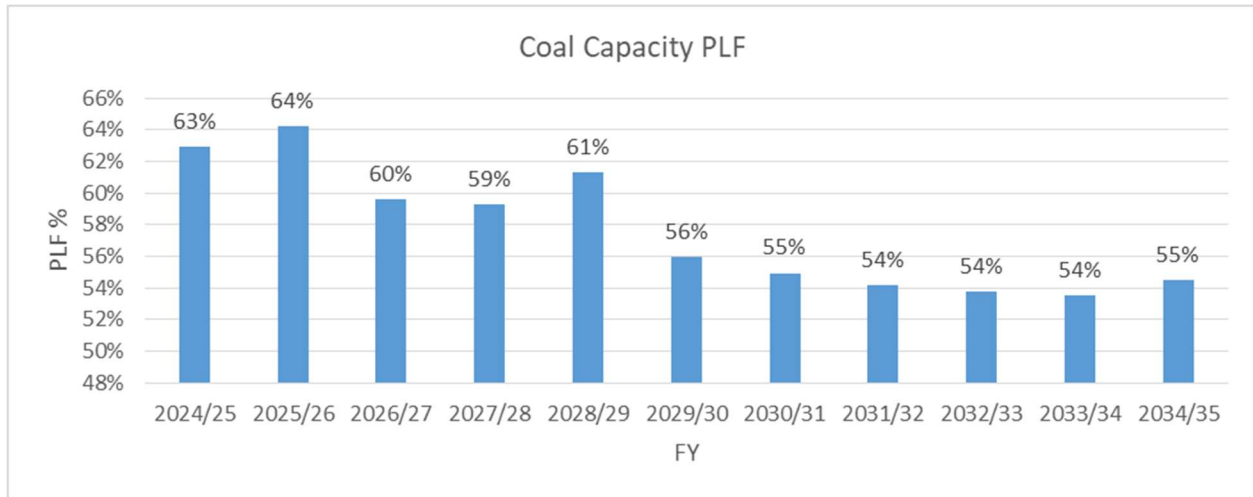


Figure 13 Year-wise coal capacity PLF for Tamil Nadu (in %)

5.5 Projected Aggregate Surplus generation for Tamil Nadu (MW)

Surplus generation is likely to be available with the state due to RE availability, Demand variation etc. The pattern of surplus capacities for Tamil Nadu has been observed as shown below in Figure 14. This capacity can be shared with other states and reduce the fixed cost burden on the utilities resulting in reduction in the cost for consumer. Tamil Nadu has likely average hourly surplus capacity available in the range of 2000-7500 MW during the months of June to September in FY 2026-27. This indicates that the state may look for banking / medium-term bilateral agreements with other states to optimize utilization of these likely surpluses.

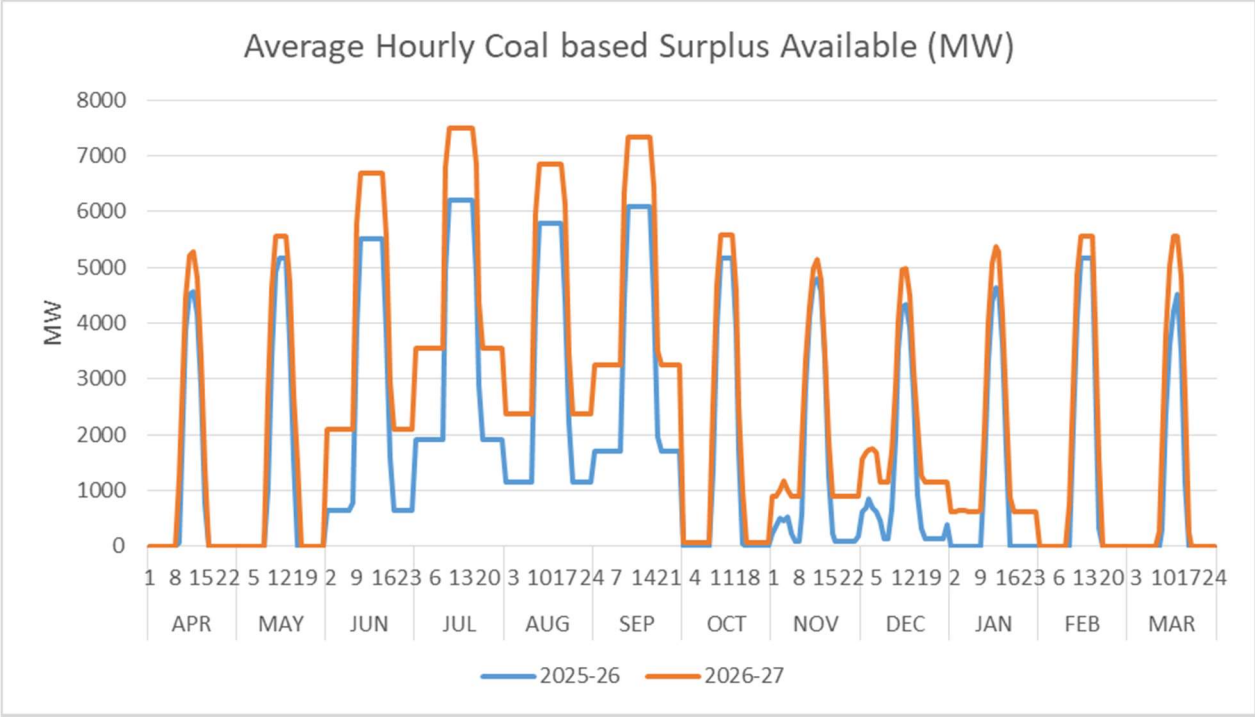


Figure 14 Average hourly Surplus Coal generation (MW) for the years 2025-26 and 2026-27

6.0 Conclusion

Based on the RA study results, the following observations are made:-

1. The study was carried out to assess the resource adequacy of Tamil Nadu based on the demand projections by Tamil Nadu (The demand projections by the 20th Electric Power Survey (EPS) are lower compared to the demand projections by Tamil Nadu).
2. The study is carried out based on the hourly load pattern of the FY 2023-24. The peak demand month is March.
3. Tamil Nadu is likely to witness energy deficit throughout the period of study i.e. from 2024-25 to 2034-35 with the existing, planned capacity including the capacity required to meet Renewable Purchase Obligations (RPO). It was observed that the total unserved energy in the year 2034-35 is likely to be around 45587 MU. Further, it is expected that the shortfall would be maximum during the non-solar hours during the months of October to April.
4. To meet the projected demand reliably, an additional 7000 MW from Coal, 5500 MW from Solar, 5500 MW from Wind, and 11680 MW from Storage may be required with a planning reserve margin of 19%. Further, the study suggests year-wise short-term/medium-term/bilateral requirements to meet the demand optimally.
5. The coal capacity PLF is projected to range between 59% and 64% until FY 2028-29. After that, it is expected to stabilize around 54% from FY 2028-29 to FY 2034-35, ensuring greater integration of renewable energy sources
6. Further, Tamil Nadu has likely surplus capacity available in the range of 2000-7500 MW in the month of June to September in FY 2026-27 which can be shared with other states/utilities.
7. The State is likely to meet the RPO targets from the projected capacity mix starting FY 2024-25.

Future Contracted/Approved Capacity Expansion and Termination (MW)

SR. NO	TIED CAPACITY NAME	TAMIL NADU'S SHARE (MW)	TYPE OF GENERATION	EXPECTED COD
1	North Chennai Thermal Power Plant (Stage III) Project (1x800 MW)	800	COAL	2024-25
2	Udangudi Thermal Power Project Stage-I (2x660 MW)	1320	COAL	2024-25
3	Ennore SEZ Supercritical Thermal Power Project (2x660 MW)	1320	COAL	2026-27
4	NTPC - Talcher Stage III	462	COAL	2027-28
5	NLC TPS II 2nd Expansion	1082	COAL	2028-29
6	Talabira Project of NLC Unit 1&2	1000	COAL	2028-29
7	Talabira Project of NLC Unit 3	500	COAL	2029-30
8	Bhavini Kalpakkam (PFBR) (1x500 MW)	151.8	NUCLEAR	2024-25
9	Kudankulam Atomic Power Station Unit 3&4	1100	NUCLEAR	2026-27
10	Kudankulam Atomic Power Station Unit 5&6	1000	NUCLEAR	2029-30
11	Kollimalai Hydro Electric Project (1x20 MW)	20	HYDRO	2025-26
12	Kundah Pumped Storage Hydro Electric Project (4x125 MW)	500	PSP	2025-26
13	Upper Bhavani	1000	PSP	2033-34

SR. NO	TIED CAPACITY NAME	TAMIL NADU'S SHARE (MW)	TYPE OF GENERATION	EXPECTED Year of PPA Termination/Retirement
1	BARH II	111	COAL	2024-25
2	Barh STPP-I	94	COAL	2024-25
3	DBPL (MTOA)	100	COAL	2025-26
4	JVPL (MTOA)	50	COAL	2025-26
5	OPG (LTOA)	74	COAL	2028-29
6	NEYVELI (EXT) TPS	193	COAL	2028-29
7	Coastal Energen	558	COAL	2028-29
8	PTC STPP	100	COAL	2028-29
9	KAMALANGA TPS	102	COAL	2028-29
10	Jindal (LTOA)	400	COAL	2028-29
11	ITPCL TPP	540	COAL	2028-29
12	EMCO WARORA TPS	150	COAL	2028-29
13	DHARIWAL TPP	100	COAL	2028-29

14	BALCO-I (LTOA)	100	COAL	2028-29
15	BALCO-II (LTOA)	100	COAL	2028-29
16	KSK Mahanadi	500	COAL	2029-30
17	DB Power (LTOA)	208	COAL	2029-30
18	TAQA	250	COAL	2033-34
19	NEYVELI TPS- II	633	COAL	2034-35
20	PPN Power	331	GAS	2031-32
21	KUDANKULAM	1178	NUCLEAR	2028-29

Assumption for Resource Adequacy Studies for the state of Tamil Nadu

1. Electricity Demand & peak requirement: As per Tamil Nadu Projections
2. Demand Profile: Based on hourly demand profile of 2023-24
3. Existing, Planned, and Expiring Capacity Due to PPA Termination/Retirement: As per the information received from TAMIL NADU
4. Future RE Capacity addition: based on RPO trajectory
5. Cost parameters: based on information in the National Electricity Plan

RE CUF considered

Hydro CUF Existing & Planned	Solar CUF Existing & Planned	Wind CUF Existing & Planned	Biomass
32%	17.8% & 22%	16.4% & 29%	21.5%

Technical Parameters

Technology	Type	Availability (%)	Ramping (%/min)	Min. Technical . (%)	Start -up time (hr)		
					Hot	Warm	Cold
Coal/ Lignite	Existing/Planned	85	1	55	2	5	10
	Candidate	88	1	55	2	5	10
Gas	Existing	90	5	40	1.5	2	3
Nuclear	Existing/Planned	68	Const. Load	-	-	-	-
Biomass	Existing/Planned	60	2	50	2	4	8
Hydro	Existing/Planned/ Candidate	As per available hourly generation profile	100	-	-	-	-
Solar	Existing/Planned		-	-	-	-	-
	Candidate		-	-	-	-	-
Wind	Existing/Planned		-	-	-	-	-
	Candidate		-	-	-	-	-
Pumped storage	Existing/Planned	95	50	-	-	-	-
	Candidate		50	-	-	-	-

Battery Energy Storage	Candidate	98	NA	-	-	-	-
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Technology	Type	Heat Rate (MCal/MWh)		Aux. Consum. (%)	Min. online time (hr)	Min. offline time (hr)	Start-up fuel consumption (MCal/MW)		
		At max loading	At min loading				Hot	Warm	Cold
Coal	Existing/ Planned	2300 to 2879	2438 to 3052	7.0	6	4	600	1000	1800
	Candidate (SC & USC)	2060 to 2125	2183 to 2253	6.5	6	4	600	1000	1800
Gas	Existing	2000 to 2900	2260 to 3277	2.5	4	3	30	50	90
Nuclear	Existing/ Planned	2777	2777	10	6	4	-	-	-
	Candidate	2777	2777	10	-	-	-	-	-
Biomass	Existing/ Planned	4200	4450	8	6	4	600	1000	1800
	Candidate	4200	4450	8	6	4	600	1000	1800
Hydro	Existing/ Planned	-	-	0.7	-	-	-	-	-
	Candidate	-	-	0.7	-	-	-	-	-
Pumped Storage	Existing/ Planned	-	-	pump efficiency	-	-	-	-	-
	Candidate	-	-	80 %	-	-	-	-	-
Battery Energy Storage	Candidate	-	-	Round trip losses 12%	-	-	-	-	-

Transmission Parameters

A single node has been considered for the purpose of study with all generating units and demand connected to the node. No transmission bottleneck has been considered for the study. Interstate ATC limit has not been considered in the study.

Financial Parameters

Following cost parameters have been assumed in the year 2021-2022:

Resource	Capex* (in ₹/MW)	O&M Fixed Cost (in ₹/MW)	Construction Time (in years)	Amortization /Life time (in years)
Coal	9.3 Cr	19.54 Lakh	4	25
Hydro~	6 Cr to 20 Cr	2.5% of Capex	5 to 8	40
Solar**	4.5 Cr to 4.1 Cr	1 % of Capex	0.5	25
Wind(Onshore)	6 Cr	1% of Capex	1.5	25
Wind(Offshore)	13.7 Cr	1% of Capex	1.5	25
Biomass	9 Cr	2% of Capex	3	20
Pumped Storage	3 Cr to 8 Cr	5 % of Capex	7	40
Battery Energy Storage (2-Hour)	5.13 Cr to 3.13 Cr	1 % of Capex	0.5	14
Battery Energy Storage (4-Hour)	8.22 Cr to 4.72 Cr	1 % of Capex	0.5	14
Battery Energy Storage (5-Hour)	9.77Cr to 5.51 Cr	1 % of Capex	0.5	14
Battery Energy Storage (6-Hour)	11.31 Cr to 6.30 Cr	1 % of Capex	0.5	14

* All the Capex figures are on actual basis at the cost level of 2021-22 i.e., inflation is not considered while calculating capex.

~ The Capex values of Hydro and PSS candidates are considered as per the project cost details furnished by the respective developers for state and private sector plants and as per RCEs done periodically by CEA for central sector plants.

**Solar Cost is assumed to reduce from Rs 4.5 Cr/MW in 2021-22 to Rs 4.1 Cr/MW in 2029-30.